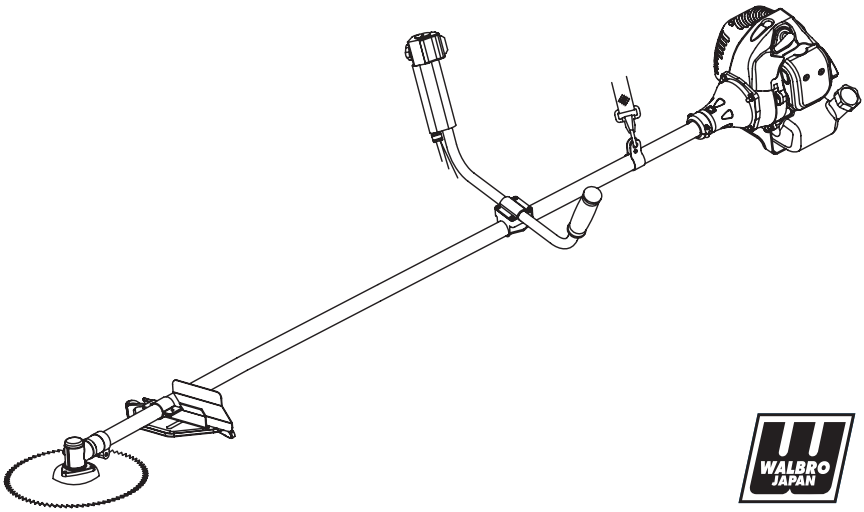
KASEI
Advanced Technology

# OPERATOR'S MANUAL

# KZ-4000TP-PRO-2S

## TEA PRUNER



**Original Instructions**

**CAUTION:** • Read all precautions and instructions in this manual before using this product. Keep this manual for future reference. • Product may vary slightly from picture.

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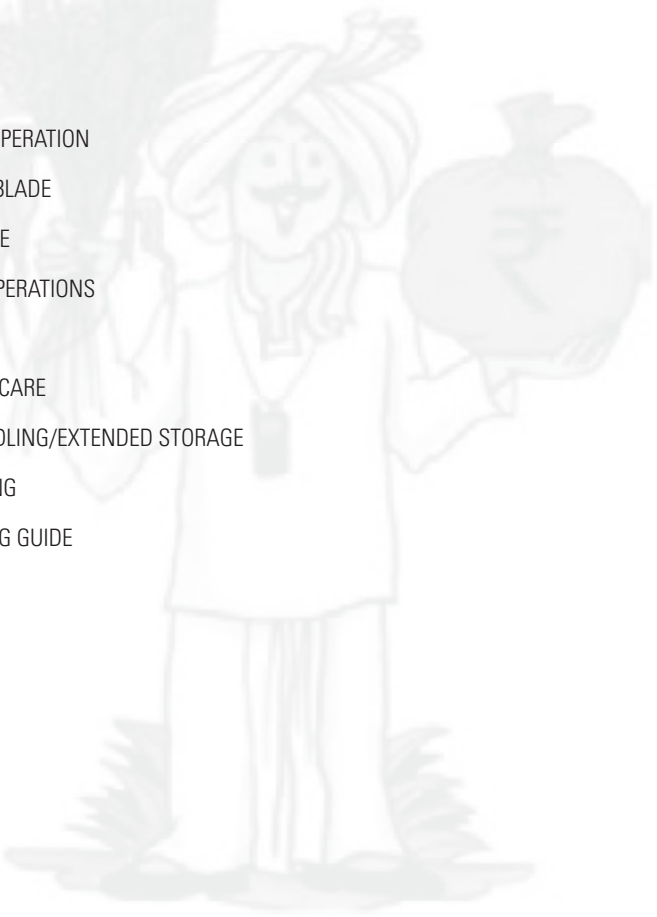
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# INTRODUCTION

Thank you for your purchasing the brush cutter of our Company. The present manual explains how to handle the brush cutter well. Before using it, please read this operator's manual carefully, operate the machine correctly and engage in your task with safely. By the way, due to changes of specifications, all details of your machine may not agree with this manual. Please understand accordingly.

**TECHNICAL DATA**

| MODEL | | KZ-4000TP-PR0-2S |
|---|---|---|
| Main unit | Remote type | Automatic centrifugal clutch; Spiral bevel gear; Drive shaft |
| | Maximum speed of gear shaft/ engine speed (r/min) | 7000/9000 |
| | Idle speed(r/min) | 2800±280 |
| | Vibration values at each handle($m/s^2$) | ≤ 9.5 |
| | Sound pressure level L (in accordance with ISO10886, 7917) | ≤ 105 |
| | Sound power level L (in accordance with ISO 10884) | ≤ 120 |
| | Reduction ratio | 16:20 |
| | Blade rotation direction | Counter-Clockwise(seen from the top) |
| | Type of handle | Double grip |
| | Weight when dry(kg) | 7.7 |
| Engine | Name of engine | HP40 |
| | Type | Air-cooled; 2 cycle; vertical piston valve; gasoline engine |
| | Displacement(ml) | 39.8 |
| | Maximum output(kW/r/min) | 1.25/7000 |
| | Carburetor | Diaphragm-type |
| | Ignition | Non-contact electronic ignition |
| | Method of starting | Recoil type |
| | Fuel used | Gasoline mixed with lubricating oil (ratio of 30:1) |
| | Fuel tank capacity(L) | 1 |
| | Dry weight(kg) | 4.2 |
| Length of main pipe section(mm) | | 1255 |
| Length of drive shaft section(mm) | | 1290 |
| Standard blades(mm) | | CG430B.3 255mm 80-blade (255x1.25) |

Technical Data subject to changed without notice.

# GRAPHIC WARNINGS

Because a trimmer/cutter is a high-speed, fast cutting power tool, special safety precaution must be observed to reduce the risk of personal injury. Read this manual carefully. Be familiar with the controls and the proper use of the unit. Know how to shut it off, and how to unhook a harnessed unit quickly.

| | |
|---|---|
|  | This graphic accompanied by the words warning and danger calls attention to an act of a condition which can lead to serious injury. |
|  | Wear safety boots |
|  | Wear safety gloves |
|  | READ & UNDERSTAND Operator's manual and safety manual. |
|  | Always wear eye and ear protection. |
|  | Wear head protection where there is a risk of falling objects |
|  | Always turn off engine and make sure the cutting tool has stopped before cleaning, removing or adjusting blade |
|  | Keep bystanders away at least 15m/50feet |
|  | **WARNING!** Never modify a trimmer or cutter in any way. improper use of any trimmer can cause serious or fatal personal injury. |
|  | **WARNING! Do not touch hard wood; stone; steel.** |

WARNING ⚠ & DANGER

The engine exhaust from this product contains chemicals to cause cancer, birth defects, of other reproductive harm.

Specifications, descriptions and illustrative material in this manual are as accurate as known at the time of publication, but are subject to change without notice. Illustrations may include optional equipment and accessories, and may not include all standard equipment

# SAFETY RULES & PRECAUTIONS

Proper safety precautions must be observed. Like all power equipment this unit must be handled carefully. Do not expose yourself or others to danger. Follow these general rules. Do not permit others to use this machine unless they are thoroughly responsible and have read, and understand the machine manual and are trained in its operation.

- Always wear safety goggles for eye protection. Dress properly , do not wear loose clothing or jewelry that could become caught in moving parts of the unit. Safe, sturdy, nonskid footwear should always be worn. Long hair should be tied back. It is recommended that legs and feet be covered to protect from flying debris during operation.
- Inspect the entire machine for loose parts (nuts, bolts, screws, etc.) and any damage. Repair or replace as necessary before using the machine.
- DO NOT USE any attachment with this power head other than the ones recommended by our company. Serious injury to the operator or bystanders could result as well as damage to the machine.
- Keep the handles away from oil and fuel.
- Always use proper handles and shoulder strap when cutting.
- Do not smoke while mixing fuel or filling tank.
- Do not mix fuel in an enclosed room or near open flames. Assure adequate ventilation. I Always mix and store the fuel in a properly marked container that is approved by local codes and ordinances for such usage.
- Never remove the fuel tank cap while the engine is running.
- Never start or run the engine inside a closed room or building. Fumes from the exhaust contain dangerous carbon monoxide.
- Never attempt to make engine adjustments while the unit is running and strapped to the operator. Always make engine adjustments with the unit resting on a flat, clear surface.
- Do not use the unit if it is damaged or poorly adjusted. Never remove the machine's guard. Serious injury to the operator or bystanders could result as well as damage to the machine.
- Inspect the area to be cut and remove all debris that could become entangled in the nylon cutting head. Also remove any objects that the unit may throw during cutting.
- Keep children away. Onlookers should be kept at a safe distance from the work area, at least 15 METER. (50 FEET)
- Never leave the machine unattended.
- Do not use this unit for any job other than those for which it is intended as described in this manual.
- Do not overreach. Keep proper footing and balance at all times. Do not run the unit while standing on a ladder or on any other unstable footing location.
- Keep hands and feet clear of the nylon cutting head while unit is in use.

- Do not use this type of machine for sweeping away debris.
- Do not use the unit when you are tired, ill or under the influence of medication, drugs or alcohol.
- Use nylon cutting head which is free of damage. If a stone or any other obstacle is hit, stop the engine and check the nylon cutting head. A broken or unbalanced nylon cutting head must never be used.

## FOLLOW INSTRUCTION FOR CHANGING ACCESSORIES.

- Do not store in a closed area where fuel vapors can reach an open flame from hot water heaters, heaters, furnaces, etc. Store in a locked, well ventilated area only.
- ensure safe and proper performance of your product. These part are available from your dealer. The use of any other accessories or attachments may cause a potential hazard or injury to the user, damage to the machine and void this warranty.
- Clean the machine completely, especially, the fuel tank, its surroundings, and the air cleaner.
- When refueling, be sure to stop the engine and confirm that it is cooled down. Never refuel when the engine is running or hot. When gasoline spills, be sure to wipe it up completely and properly dispose of those materials before starting the engine.
- Stay clear of other workers or bystanders by at least 15 meters /50 feet.
- When ever approaching an operator of the machine, carefully call his attention and confirm that the operator stops the engine. Be careful not to startle or distract the operator which can cause an unsafe situation.
- Never touch the nylon cutting head when ever the engine is running. If it is necessary to adjust the protector or nylon cutting head, be sure to stop the engine and confirm that the nylon cutting head has stopped running.
- The engine should be turned off when the machine is moved between work areas.
- Be careful not to hit the nylon cutting head against stones, or the ground. Unreasonable rough operation will shorten the life of the machine as well as create an unsafe environment for yourself and those around you.
- Pay attention to loosening and overheating of parts. If there is any abnormality of the machine, stop operation immediately and check the machine carefully. If necessary, have the machine serviced by a authorized dealer. Never continue to operate a machine which may be malfunctioning.
- In start-up or during operation of the engine, never touch hot parts such as the muffler, the high voltage wire or the spark plug.
- After the engine has stopped, the muffler is still hot. Never place the machine in any places where there are flammable materials (dry grass, etc.), combustible gasses or combustible liquids.
- Pay special attention to operation in the rain or just after the rain as the ground may be slippery.
- If you slip or fall to the ground or into a hole, release the throttle lever immediately.
- Be careful not to drop the machine or hit it against obstacles.

- Before proceeding to adjust or repair the machine, be sure to stop the engine and detach the spark plug cap from the spark plug.
- When the machine is placed in storage for a long time, drain fuel from the fuel tank and carburetor, clean the parts, move the machine to a safe place and confirm that the engine is cooled down.
- Make periodic inspections to assure safe and efficient operation. For a thorough inspection of your machine, please contact a dealer.
- Keep the machine away from fire or sparks.
- It is believed that a condition called Raynaud's phenomenon, which affects the fmgers of certain individuals may be brought about by exposure to vibration and cold. Exposure to vibration and cold may cause tingling and burning sensations followed by loss of color and numbness in the fingers. The following precautions are strongly recommended because the minimum exposure which might trigger the ailment is unknown.

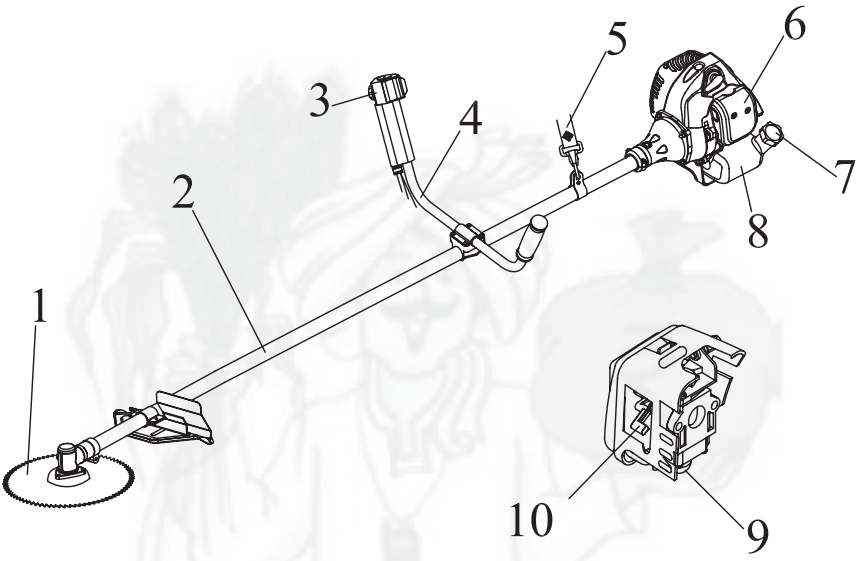
  Keep your body warm, especially the head, neck, feet, ankles, hands and wrists.

  Maintain good blood circulation by performing vigorous arm exercises during frequent work breaks and also by not smoking.

  Limit the hours of operation. Try to fill each day with jobs where operating the brush cutter or other hand-held power equipment is not required.

  If you experience discomfort, redness and swelling of the fmgers followed by whitening and loss of feeling, consult your physician before further exposing yourself to cold and vibration.
- Always wear ear protection, loud noise for long time can make hearing debase even lost hearing. Total face and head protection must be wear to prevent damage from plummet or branch.
- Wear no-slip heavy duty work gloves to improve your grip on the brush cutter handle. Gloves also reduce the transmission of machine vibration to your hands.
- **WARNING:** The cutter area is still dangerous while the machine is coasting to a stop.

# DESCRIPTION




SAFETY MANUAL included with unit. Read before operating and keep for future reference to learn proper, safe operating techniques.

1. BLADE Circular blade for grass, weed or brush cutting applications. Harness, metal shield and U-Handles required for blade operation.
2. DRIVE SHAFT ASSEMBLY Contains a specially designed liner flexible drive shaft.
3. IGNITION SWITCH "slide switch"mounted on top of the Throttle Trigger Housing.Move switch FORWARD to Stop, BACK to Run.
4. U-HANDLE this handle can be adjusted at you convenience for comfort.
5. HARNESS ASS'Y a attachment which hang the machine on the operator and adjustable in length.
6. AIR CLEANER COVER Contains replaceable felt element.
7. FUEL TANK LID.
8. FUEL TANK Contains fuel and fuel filter.
9. PURGE BULB Pushing purge lulb(7 to 10 times before starting), will remove all air and stale fuel from fuel system and bring fresh fuel to the carburetor.
10. CHOKE the choke control is located on the top of the filter cover.

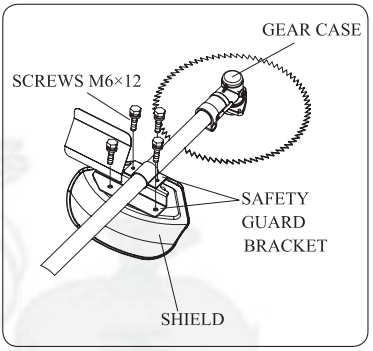
COLD STARTS Pull lever up to close choke for cold starts. Push lever down for RUN position.

# ASSEMBLY

INSTALLING PLASTIC SHIELD

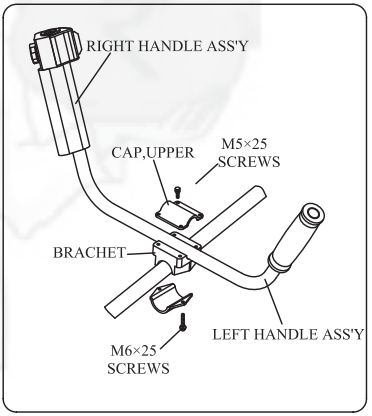
Install the safety cover on the shaft with the safety cover bracket and the screws provided.

Tighten the screws after setting the safety cover bracket at the appropriate position.




INSTALLING HANDLE

Install handle in bracket. Position handle in a comfortable operating position and tighten four M5xl6screws.

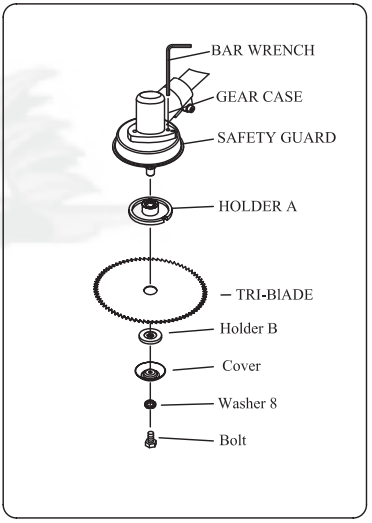



BLADE INSTALLATION

Install holder A, the tri-blade, holder B, washer 8,in this order, then clamp with a lefthand thread bolt.

Align hole of gear case and holder A, and insert S=4mm bar wrench.

Rotate blade-fastening bolt with plug box wrench in clockwise direction until securely tightened.




## WARNING ⚠ CAUTION

EXCESSIVE VIBRATION
STOP ENGINE IMMEDIATELY

Excessive vibration of blade means that it is not properly installed. Stop engine immediately and check blade. Blade improperly installed will cause injury. Use only our company original cutting equipment parts when servicing the unit.

# RULES FOR SAFE OPERATION

## WARNING ⚠ DANGER

All models trinuners and brushcutters can throw

small stone , metal or small objects as well as the material being cut. Read these"Rules for Safe Operation"With care.

Follow the instructions in your operator's manual.

Read the operator's manual carefully. Be thoroughly familiar with the controls and proper use of the unit. Know how to stop the unit and shut off the engine. Know how to unhook a harnessed unit quickly.

Never allow anyone to use the unit without proper inslruction.Be sure the operator wears the foot, leg, eye, face and hearing protection. Do not rely on the debris shield on the unit to protect your eyes from ricocheting or thrown objects. Keep the area clear of bystanders, children and pets. Never allow children to oper ate or play with the unit.

Do not allow anyone to enter the operating

DANGER ZONE with you. The danger zone the danger zone wear eye protection from thrown objects. If the unit must be used where there are unprotected people, operate at a low throttle speed to reduce the risk of thrown objects.

Move the stop switch to "stop" position when the engine idles, insure it reliable before cutting. Only may continue to operate the unit when stop switch is in good condition.

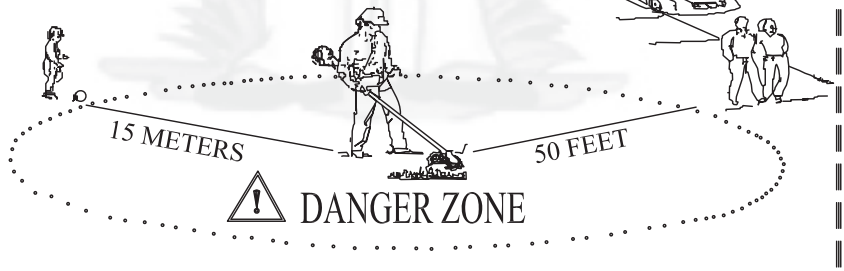
Routines for checking that the cutting attachment stops turning when the engine idles.

## WARNING ⚠ DANGER

In addition to head, eye, face and ear protectors shoes to protect your feet improve your footing on slippery surfaces. Do not wear ties, jewelry, or loose, dangling clothing which could be caught in the unit.

Do not wear open-toed footwear, or go bare foot or bare legged. In certain situations you must wear total face and head protection.

Pay attention not to kick back and blade thrust.




NO PERSON EXCEPT OPERATOR ALLOWED 15 METERS RADIUS DANGER ZONE.EYE,EAR,FACE, FOOT,LEG AND BODY PROTECTION MUST BE WORN BY OPERATOR.PERSONS IN ZONE OR RISK BEYOND DANGER ZONE SHOULD WEAR EYE PROTECTION AGAINST THROWN OBJECTS.RISK DIMINISHES WITH DISTANCE FROM DANGER ZONE.

# CUTTING WITH A BLADE

SCYTIITNG WEEDS. This is cutting by swinging the blade in a level arc. It can quickly clear areas of field grass and weeds. Scything should not be used to cut large, tough weeds or woody growths.

> NOTE
>
> Do not use a brush blade to cut trees which exceed a diameter of 112 to 3/4 inches.

Scything can be done in both directions, or just in one way which results in the debris being thrown away from you.

That is using the side of the blade that it is rotating away from you. Tilt down the blade slightly on

this same side .You will be hit by some debris if you scythe in both directions.

SAWING This must be done with a saw blade when the weeds are too thick and strong for scything.

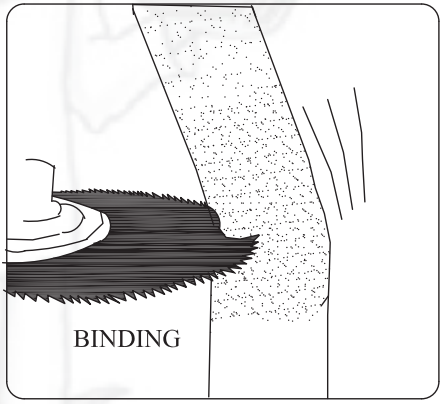
If kickback occurs, the blade may be dull, or the material so thick and hard that you should use a saw blade instead of the brush blade.

Growth up to 3-inches in diameter, which is too thick for scything, may be cut by sawing. Sawing requires judgement on your part whether to cut on the side where the blade pulls away from you but throws back the debris ,or on the side where the debris is thrown away from you but the reaction is push which could become a kickout(blade thrust).

Always cut on the side of the tree which will cause the tree to fall away from the unit. A kickout or bind may result if the tree falls toward the unit and pinches the blade.

Do not force the blade to cut. Do not change the angle of cut after into the wood. Avoid using any pressure or leverage which could cause the blade to bind or crack.


SAWING SAPLINGS



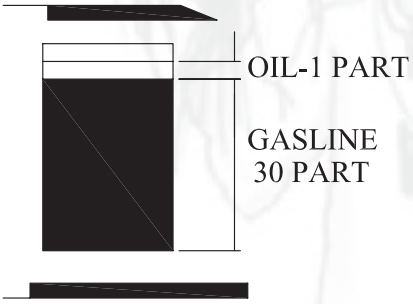



# FUEL & OIL MIXTURE

WARNING

- Never fill the fuel tank to the very top.
- Never add fuel to the tank in a closed non-ventilated area.
- Do not add fuel to this unit close to an open fire or sparks.
- Be sure to wipe off spilled fuel before attempting to start engine.
- Do not attempt to refuel a hot engine.

Fuel used for this model is a mixture of unleaded gasoline and approved engine lubricant. When mixing gasoline with two-cycle engine oil, use only gasoline which contains NO ETHANOL or METHANOL (Types of Alcohol) Use Branded 89 octane or higher unlead gasoline known to be of good quality. This will help to avoid possible damage to engine fuel lines and other engine parts.




MIXTURE RATIO IS 30:1

GASLINE-30PART OIL-1 PART

Fuel mixture at the rate other than 30:1 may cause damage to the engine,Ensure mixture ratio is correct.

FUEL

The engine uses two-stroke fuel, a mixture of gasoline and 2-stroke lubricant 30:1

GASOLINE

Use branded 89 octane or higher unleaded gasoline known to be of good quality.

STORING FUEL

Store fuel only in a clean, safe, approved container. Check and follow local ordinances on type and location of storage container.

IMPORTANT

Two-stroke fuel may separate.Shake fuel container thoroughly before each use. Stored fuel ages. Do not mix more fuel than you expect to use within a month.

FUEL AND OIL MIXTURE

Inspect fuel tank making sure that it is clean and fill with fresh fuel. Use a mixture of30-1

# PREPARING FOR OPERATIONS

When mixing gasoline with two cycle engine oil, use only gasoline which contains no ethanol or methanol(types of alcohol). this will help to avoid possible damage to engine fuel lines and other engine parts.

Do not mix gasoline and oil directly in the engine fuel tank.

IMPORTANT:Failure to follow proper fuel mix instructions may cause damage to the engine.

I. When preparing fuel mixture, mix only the amount needed for the job you are to do. Do not use fuel that has been stored longer than two months.Fuel mixture stored longer than this will cause hard starting and poor performance.If fuel mix has been stored longer than this time, it should be removed and filled with fresh mixture.

## WARNING ⚠ DANGER

2. Never fill the fuel tank to the very top.(under 3/4 of the tank)
3. Never add fuel to the tank in a closed non-ventilated area.
4. Do not add fuel to this unit close to an open fire or sparks.
5. Be sure to wipe off spilled fuel before attempting to start engine
6. Do not attempt to refuel a hot engine.

## CHECK POINTS BEFORE OPERATION

I. Check for loose bolts, nuts and fittings.
2. Check the air cleaner for dirt. Clean the air filter of all dirt, etc. before operation.
3. Check to be sure that protector is securely in place.
4. Check to be sure that fuel is not leak.
5. Check to be sure that blade is no crack.

## WARNING ⚠ DANGER

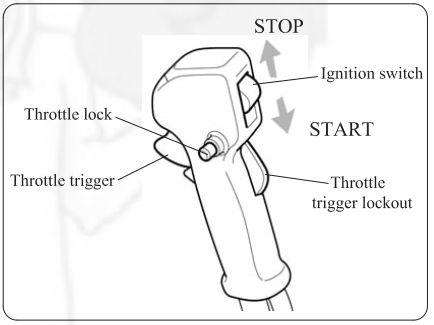
The engine exhaust from this product contains chemicals known to cause cancer, birth defects, or other reproductive harm.

# OPERATIONS

## ENGINE STARTING AND STOPPING PROCEDURES STARTING COLD ENGINE

1. Move the ignition switch to "START" position.
2. Give a gentle push on the primer pump repeatedly (7-10times) until fuel comes into the primer pump.
3. Pull the chock lever up to close choke
4. Move the throttle trigger and keep it in position with the throttle lock (if present).
5. Pull starter handle until engine false fires.
6. Push choke lever inwards( excessive cranking with choke lever will cause flooding of engine making it difficult to start.)
7. Pull starter handle until engine starts.
8. When the engine is started, disconnect the starter by turning the choke lever to "RUN"
9. Disengage the throttle lock (if present)pressing the throttle trigger briefly.
10. Allow engine to warm up for a few minutes before using.

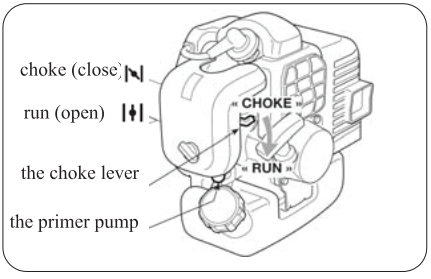



## STARTING WARM ENGINE

The choke lever to the "RUN"

- If fuel tank was not run dry, pull starter one to three times and engine should start.
- If fuel tank was run dry, after refilling repeat steps 1-2-7 .




*If the starter rope is pulled repeatedly with the choke on, it may flood the engine and make starting difficult.*

If you have flooded the engine, remove the spark plug and gently pull the handle on the starter rope to eliminate any excess fuel; then dry the spark plug electrodes and replace it on the engine.

## STOPPING THE ENGINE

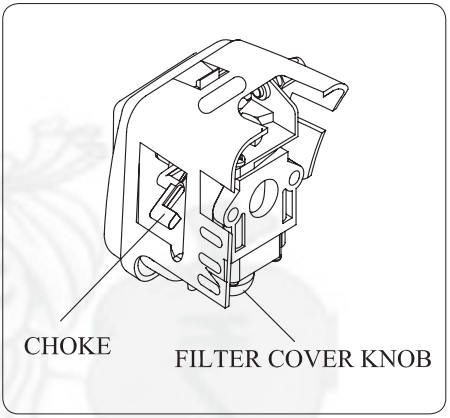
- Release the throttle trigger and allow the engine to run idle for a few seconds.
- Move switch to "STOP" position.

# MAINTENANCE & CARE

## AIR FILTER

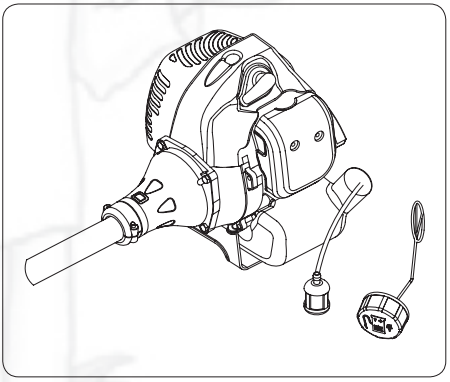
- Accumulated dust in the air filter will reduce engine efficiency. Increase fuel consumption and allow abrasive particles to pass into the engine.
- Remove the air filter as often as necessary to maintain in a clean condition.

  Light surface dust can readily be removed by tapping the filter. Heavy deposits should be washed out in suitable solvent.
- Remove filter cover by loosening air filter cover knob.




## ADJUSTING CARBURETOR NOTE

Do not adjust carburetor unless necessary. If you have trouble with the carburetor, see your dealer. Improper adjustment may cause engine damage and void warranty.



## FUEL FILTER

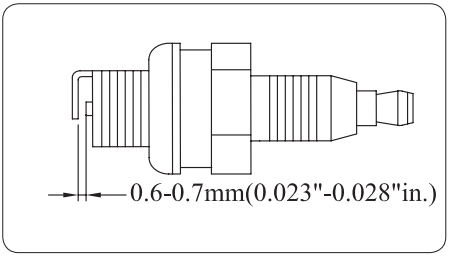
- Fuel tank is fitted with a filter.

  Filter is situated at the free end of fuel pipe and can be picked out through fuel port with a piece of hooked wire or the like.

  Check the fuel filter periodically. Do not allow dust to enter into fuel tank. Clogged filter will
- cause difficulty in starting engine or abnormalities in engine performance.
- When filter is dirty, replace the filter.
- When the inside of the fuel tank is dirty, it can be cleaned by rinsing the tank out with gasoline.

## CHECK SPARK PLUG

- Do not attempt to remove the plug from a hot engine in order to avoid possible damage to the threads.



# TROUBLE SHOOTING

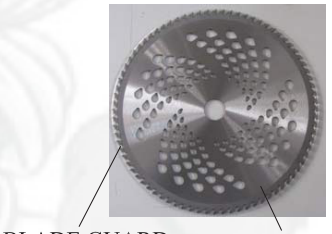
- Clean or replace the plug if fouled with heavy oily deposits.
- Replace the plug if the center electrode is worn rounded at the end.
- Spark gap 0.6-0.7mm(.023"028")
- Fastening torque= 145-155kg.cm(125-135in.lb)

## TRANSPORT, HANDLING

- Blade guard should be installed when transport, handing of the unit.

## INSTALLING BLADE GUARD

- Clefte the blade guard and install on the blade ,then put the pin into the guard notch.
- The engine should be turned off when the unit is moved between work areas.
- After the engine has stopped, the muffler is still hot. never touch hot Parts such as the muffler.
- Confirm that the fuel is not leaked from tank.




INSTALLING BLADE GUARD

## EXTENDED STORAGE

- Inspect, clean and repair unit if necessary.
- Remove all fuel from tank.
- Start engine-This will consume all fuel in fuel line and carburetor.
- Remove spark plug and pour one tea spoon of clean motor oil into spark plug hole of cylinder-replace spark plug.
- Store in clean, dry, dust free area.

FAILURES such as difficulty in starting engine, irrgularity in functions and abnormality in performance can normally be prevented if careful attention is paid to all operating instructions and maintenance procedures. Should the engine not funtion properly, check the following:

- That the fuel system is in good condition and fuel is flowing to the engine.
- The ignition system is in good condition and the spark plug, sparks correctly.
- Compression of the engine is adequate.

## FAILURE TO START

- Dispose of fuel in tank.
- Insure that fuel is fresh and clean.
- Check fuel filter to make sure it is clean. (Replace filter if necessary).
- Make sure air filter elements are clean.
- Install new, properly gapped spark plug.
- When there is serious trouble with the unit, do not try to repair it yourself, but have your distributor or dealer do it for you.

# TROUBLESHOOTING GUIDE

## 1. FAILURE TO START

| symptom | | | | probable cause | remedy |
|---|---|---|---|---|---|
| cylinder compress pressure normal | spark plug spark normal | fuel system abnormality | | these is no fuel in tank<br>fuel filter obstruct | add fuel in tank<br>clean fuel filter |
| | | fuel system normal | fuel | fuel is too dirty<br>there is water in fuel<br>there is too much fuel in cylinder<br>mixture ratio is improper | instead fuel<br>instead fuel<br>tear down spark plug and dry it<br>mixture proration |
| | fuel system normal | high voltage wrre spark normal | spark plug | spark plug fouled with oily deposits<br>spark plug insulation damage<br>spark gap is too large or smal | clean the oily deposits<br>replace spark plug<br>adjust spark gap 0.6-0.7mm |
| | | high voltage wrre spark abnormality | | high voltage wire breach or breakoff<br>coil looseness | replace or tighten<br>tighten |
| fuel system normal | ignition system normal | compress press IS shortage | | piston ring attrite<br>piston ring is breaked<br>piston ring cementation<br>spark plug looseness<br>conjoint surface of the cylinder and crank cass leak | replace a new<br>replace<br>eliminate<br>tighten<br>eliminate |
| | | compress normal | | high voltage wire and sparkplug contact badness<br>stop switch failure or short circuit | tighten the spark plug cap<br>repair or replace |

# TROUBLESHOOTING GUIDE

## 2. LOW OUTPUT

| symptom | probable cause | remedy |
|---|---|---|
| flameout when speedup<br><br>the smoke is thin, carburetor spout backward | fuel filter obstruct, fall short of fuel<br><br>muffler fouled with oily deposit | clean fuel filter, clean fuel road adjust carburetor<br><br>clean the oily deposits |
| compress press is shortage | piston, piston ring, cylinder attrite, | replace piston, piston ring |
| engine leak | conjoint surface of the cylinder and crank cass leak | reparr |
| the end of two crank shaft leak | the seal is bad | replace seal |
| | engine overheats<br><br>burning room fouled with oily deposits | avoid use it for long time, high speed and heavy load clean the oily deposits |

## 3. ENGINE RUNNING UNSTABLE

| symptom | probable cause | remedy |
|---|---|---|
| there are knock sound in engine | piston, piston ring, cylinder attrite,<br><br>piston pin, piston attrite bearing of crank shaft attrite | replace piston, piston ring<br><br>replace piston pin, piston replace bearing |
| there are metal knock sound | engine overheats<br><br>burning room fouled with oily deposits<br>gasoline branded is unfit | avoid use it for long time , high speed and heavy load clean the oily deposits<br><br>replace require branded octane |
| engine ignition break off | there is water in fuel<br>spark gap is wrong coil gap is wrong | instead fuel<br>adjust spark gap 0.6-0.7mm<br>adjust coil gap 0.3-0.4mm |

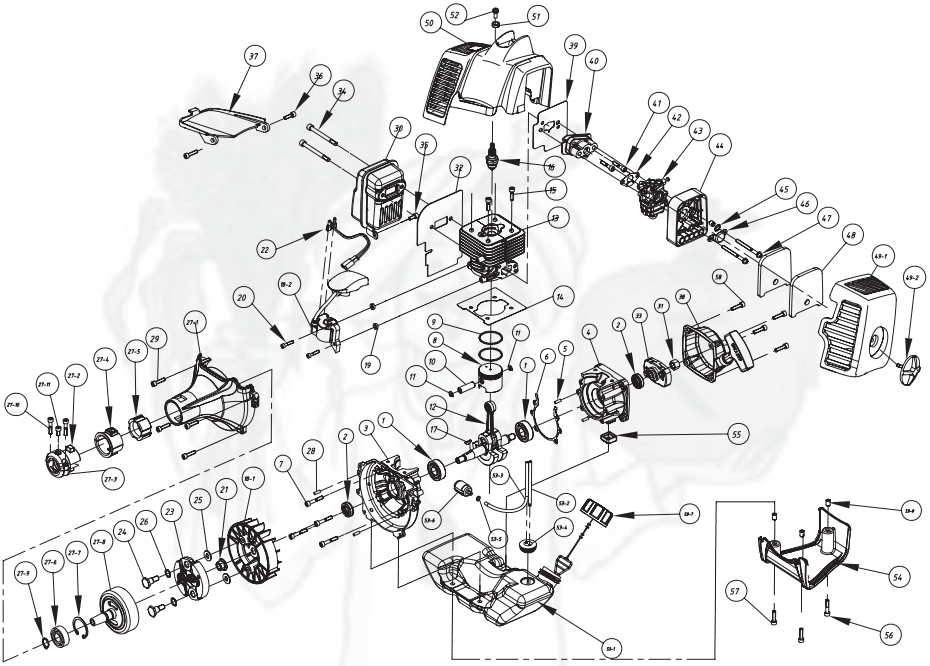
# TROUBLESHOOTING GUIDE

4. THE ENGINE SUDDENLY STOPPED

| probable cause | remedy |
|---|---|
| fuel is use up | add fuel |
| spark plug fouled with oily deposits and short circuit high voltage wire fall off | clean the oily deposits connect |



# ENGINE DIAGRAM OF KASEI-TEA PRUNER-KZ-4000TP-PRO-2S

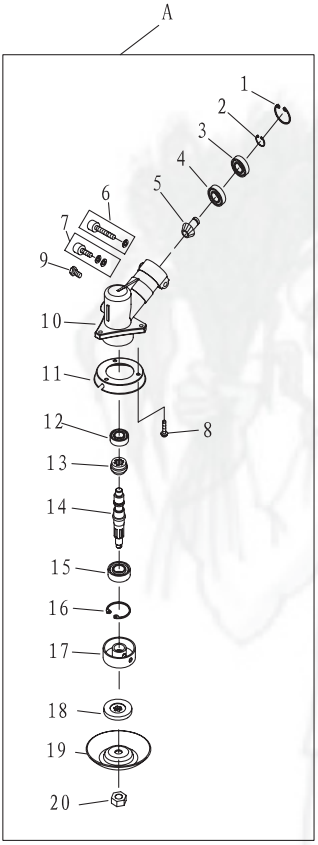


# ENGINE SPARE PARTS LIST OF KASEI-TEA PRUNER-KZ-4000TP-PRO-2S

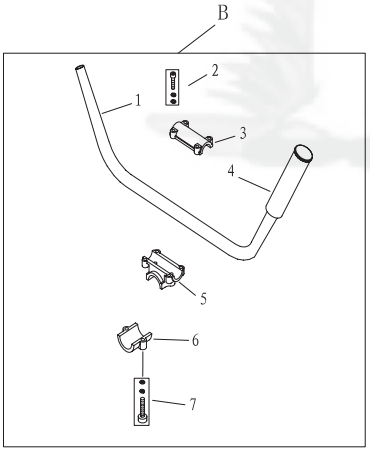
| S.N. | Part No. | Part Name |
|---|---|---|
| 1 | KZ-4000TP-E1 | Bearing 6202/P4 |
| 2 | KZ-4000TP-E2 | Oil seal |
| 3 | KZ-4000TP-E3 | Crank case F |
| 4 | KZ-4000TP-E4 | Crank case R |
| 5 | KZ-4000TP-E5 | Pin 4 x h8 x 10 |
| 6 | KZ-4000TP-E6 | Gasket crank case |
| 7 | KZ-4000TP-E7 | Screw M5 x30 |
| 8 | KZ-4000TP-E8 | Piston |
| 9 | KZ-4000TP-E9 | Ring piston |
| 10 | KZ-4000TP-E10 | Pin piston |
| 11 | KZ-4000TP-E11 | Ring snap |
| 12 | KZ-4000TP-E12 | Crank shaft assy |
| 13 | KZ-4000TP-E13 | Cylinder |
| 14 | KZ-4000TP-E14 | Gasket cylinder |
| 15 | KZ-4000TP-E15 | Screw M5 x18 |
| 16 | KZ-4000TP-E16 | Spark plug |
| 17 | KZ-4000TP-E17 | Key 3x13x5 |
| 18-1 | KZ-4000TP-E18-1 | Rotor magnet |
| 18-2 | KZ-4000TP-E18-2 | IG coil |
| 19 | KZ-4000TP-E19 | Spacer Ig coil |
| 20 | KZ-4000TP-E20 | Screw M4 x20 |
| 21 | KZ-4000TP-E21 | Nut MIOxl. 25 |
| 22 | KZ-4000TP-E22 | Wire lead assy |
| 23 | KZ-4000TP-E23 | Clutch assy |
| 24 | KZ-4000TP-E24 | Clutch bolt |
| 25 | KZ-4000TP-E25 | Clutch washerB |
| 26 | KZ-4000TP-E26 | Wave washer |
| 27-1 | KZ-4000TP-E27-1 | Fan cover |
| 27-2 | KZ-4000TP-E27-2 | Holder A |
| 27-3 | KZ-4000TP-E27-3 | Holder B |
| 27-4 | KZ-4000TP-E27-4 | Rubber A |
| 27-5 | KZ-4000TP-E27-5 | Rubber B |
| 27-6 | KZ-4000TP-E27-6 | Bearing 6202-2RZ/P5 |
| 27-7 | KZ-4000TP-E27-7 | Circlips for holes-type 1 IS |
| 27-8 | KZ-4000TP-E27-8 | Clutch drum assy 78 |
| 27-9 | KZ-4000TP-E27-9 | Circlips for shift-type 1 li |
| 27-10 | KZ-4000TP-E27-10 | Screw M5 x 20 |
| 27-11 | KZ-4000TP-E27-11 | Screw M5 x10 |
| 28 | KZ-4000TP-E28 | Pin 4h8x10 |

| S.N. | Part No. | Part Name |
|---|---|---|
| 29 | KZ-4000TP-E29 | Screw M5 x 20 |
| 30 | KZ-4000TP-E30 | Muffler ass'y |
| 31 | KZ-4000TP-E31 | Nut MIOxl. 25 |
| 32 | KZ-4000TP-E32 | Gasket muffler |
| 33 | KZ-4000TP-E33 | Guide Cover Ass'y |
| 34 | KZ-4000TP-E34 | Screw M6 x 70-10. 9 |
| 35 | KZ-4000TP-E35 | Screw M5 x 12 |
| 36 | KZ-4000TP-E36 | Screw M5 x 16 |
| 37 | KZ-4000TP-E37 | Clapboard |
| 38 | KZ-4000TP-E38 | Starter |
| 39 | KZ-4000TP-E39 | Gasket insulator |
| 40 | KZ-4000TP-E40 | Insulator |
| 41 | KZ-4000TP-E41 | Screw M5 x 25 |
| 42 | KZ-4000TP-E42 | Gasket carburetor |
| 43 | KZ-4000TP-E43 | Carburetor |
| 44 | KZ-4000TP-E44 | Cleaner body |
| 45 | KZ-4000TP-E45 | Wave washer 8 |
| 46 | KZ-4000TP-E46 | Spit back shield |
| 47 | KZ-4000TP-E47 | Screw M5 x 55 |
| 48 | KZ-4000TP-E48 | Air filter element |
| 49-1 | KZ-4000TP-E49-1 | Cover Cleaner |
| 49-2 | KZ-4000TP-E49-2 | Knob cover |
| 50 | KZ-4000TP-E50 | Top cover |
| 51 | KZ-4000TP-E51 | Gasket |
| 52 | KZ-4000TP-E52 | Screw M5 x 20 |
| 53-1 | KZ-4000TP-E53-1 | Fuel tank |
| 53-2 | KZ-4000TP-E53-2 | Pipe return |
| 53-3 | KZ-4000TP-E53-3 | Pipe intake |
| 53-4 | KZ-4000TP-E53-4 | Grommet |
| 53-5 | KZ-4000TP-E53-5 | Clip pipe |
| 53-6 | KZ-4000TP-E53-6 | Filter Assy |
| 53-7 | KZ-4000TP-E53-7 | Tank cap assy |
| 53-8 | KZ-4000TP-E53-8 | Collar C |
| 54 | KZ-4000TP-E54 | Engine stand |
| 55 | KZ-4000TP-E55 | Spacer fuel tank |
| 56 | KZ-4000TP-E56 | Screw M5 x 25 |
| 57 | KZ-4000TP-E57 | Screw M5 x 20 |
| 58 | KZ-4000TP-E58 | Screw M5 x 20 |

# KZ-4000TP-PRO-2S TRANSMISSION DIAGRAM & SPARE PARTS LIST

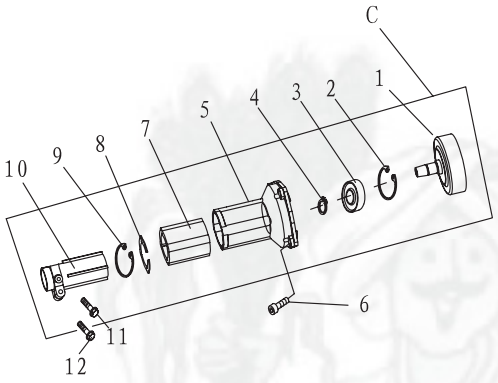



| S.N. | PART NO. | PART NAME |
|---|---|---|
| A | KZ-4000TP-AT | Gear Case Ass' y |
| 1 | KZ-4000TP-AT1 | Stop Ring 28 |
| 2 | KZ-4000TP-AT2 | Stop Ring 12 |
| 3 | KZ-4000TP-AT3 | Bearing 6001-2RS/P6 |
| 4 | KZ-4000TP-AT4 | Bearing 6001/P6 |
| 5 | KZ-4000TP-AT5 | Pinion |
| 6 | KZ-4000TP-AT6 | Screw M5X20 |
| 7 | KZ-4000TP-AT7 | Screw M5X 12 |
| 8 | KZ-4000TP-AT8 | Screw M5X 12 |
| 9 | KZ-4000TP-AT9 | Bolt M5X12 |
| 10 | KZ-4000TP-AT10 | Gear Case |
| 11 | KZ-4000TP-AT11 | Safety Guard |
| 12 | KZ-4000TP-AT12 | Bearing 6000/P6 |
| 13 | KZ-4000TP-AT13 | Gear |
| 14 | KZ-4000TP-AT14 | Gear Shaft |
| 15 | KZ-4000TP-AT15 | Bearing 6002-2RS/P6 |
| 16 | KZ-4000TP-AT16 | Stop Ring 32 |
| 17 | KZ-4000TP-AT17 | Holder A |
| 18 | KZ-4000TP-AT18 | Holder B |
| 19 | KZ-4000TP-AT19 | Bolt Cover |
| 20 | KZ-4000TP-AT20 | Left Nut |

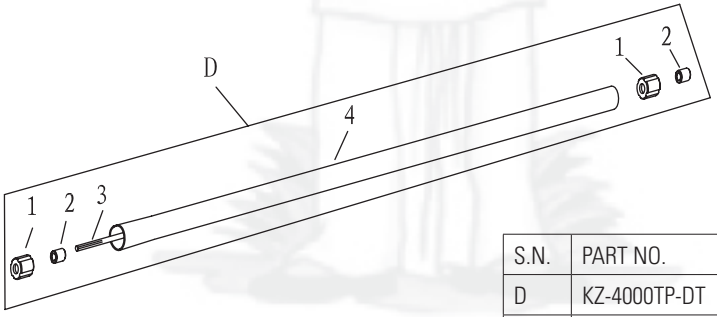



| S.N. | PART NO. | PART NAME |
|---|---|---|
| B | KZ-4000TP-BT | |
| 1 | KZ-4000TP-BT1 | Handle |
| 2 | KZ-4000TP-BT2 | Bolt M5X25 |
| 3 | KZ-4000TP-BT3 | Cap, Upper |
| 4 | KZ-4000TP-BT4 | Grip Left |
| 5 | KZ-4000TP-BT5 | Bracket |
| 6 | KZ-4000TP-BT6 | Cap, Lower |
| 7 | KZ-4000TP-BT7 | Screw M6X25 |

## KZ-4000TP-PRO-2S TRANSMISSION DIAGRAM & SPARE PARTS LIST

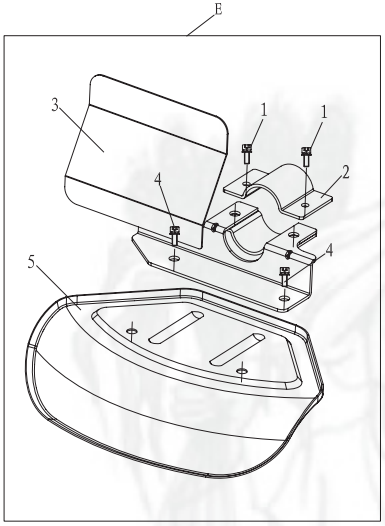



| S.N. | PART NO. | PART NAME |
|---|---|---|
| C | KZ-4000TP-CT | Clutch Comp. |
| 1 | KZ-4000TP-CT1 | Clutch Drum Comp. |
| 2 | KZ-4000TP-CT2 | Stop Ring 35 |
| 3 | KZ-4000TP-CT3 | Bearing 6202-2RS/P5 |
| 4 | KZ-4000TP-CT4 | Stop Ring 15 |
| 5 | KZ-4000TP-CT5 | Clutch Case |
| 6 | KZ-4000TP-CT6 | Bolt M6X20 |
| 7 | KZ-4000TP-CT7 | Rubber |
| 8 | KZ-4000TP-CT8 | Washer |
| 9 | KZ-4000TP-CT9 | Stop Ring 48 |
| 10 | KZ-4000TP-CT10 | Linker |
| 11 | KZ-4000TP-CT11 | Bolt M6X30 |
| 12 | KZ-4000TP-CT12 | Bolt M5X12 |

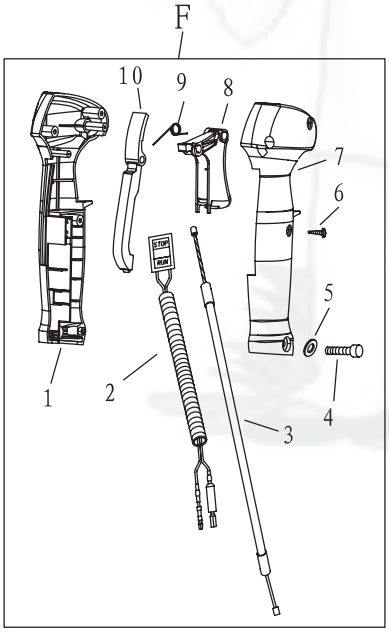



| S.N. | PART NO. | PART NAME |
|---|---|---|
| D | KZ-4000TP-DT | Pipe Comp. |
| 1 | KZ-4000TP-DT1 | Rubber Cover |
| 2 | KZ-4000TP-DT2 | Oil-bearing |
| 3 | KZ-4000TP-DT3 | Drive Shaft |
| 4 | KZ-4000TP-DT4 | Pipe |

## KZ-4000TP-PRO-2S TRANSMISSION DIAGRAM & SPARE PARTS LIST

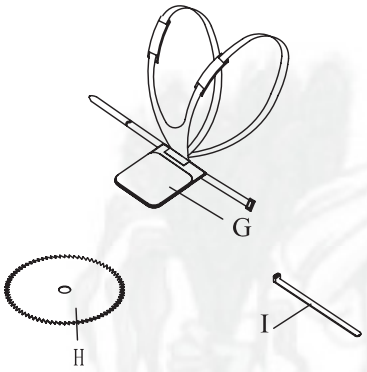



| S.N. | PART NO. | PART NAME |
|---|---|---|
| E | KZ-4000TP-ET | Safety Guard Comp. |
| 1 | KZ-4000TP-ET1 | Screw M6x20 |
| 2 | KZ-4000TP-ET2 | Safety Guard Press Plank |
| 3 | KZ-4000TP-ET3 | Safety Guard Bracket |
| 4 | KZ-4000TP-ET4 | Screw M6xl6 |
| 5 | KZ-4000TP-ET5 | Safety Guard |

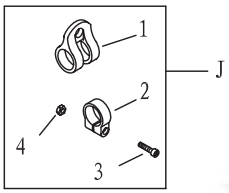



| S.N. | PART NO. | PART NAME |
|---|---|---|
| F | KZ-4000TP-FT | Lever Ass'y |
| 1 | KZ-4000TP-FT1 | Box, Left |
| 2 | KZ-4000TP-FT2 | Stop Button Comp. |
| 3 | KZ-4000TP-FT3 | Cable Comp. |
| 4 | KZ-4000TP-FT4 | BOLT M5*30 |
| 5 | KZ-4000TP-FT5 | WASHER 5 |
| 6 | KZ-4000TP-FT6 | BOLT ST3.5*13 |
| 7 | KZ-4000TP-FT7 | Box, Right |
| 8 | KZ-4000TP-FT8 | Handgrip |
| 9 | KZ-4000TP-FT9 | Spring |
| 10 | KZ-4000TP-FT10 | HANDGRIP |

## KZ-4000TP-PRO-2S TRANSMISSION DIAGRAM & SPARE PARTS LIST




| S.N. | PART NO. | PART NAME |
|---|---|---|
| G | KZ-4000TP-GT | Harness Ass'y |
| H | KZ-4000TP-HT | Blade |
| I | KZ-4000TP-IT | Wire Clamp Band |




| S.N. | PART NO. | PART NAME |
|---|---|---|
| J | KZ-4000TP-JT | Holder Ass'y |
| 1 | KZ-4000TP-JT1 | Harness Clamp |
| 2 | KZ-4000TP-JT2 | Clamp |
| 3 | KZ-4000TP-JT3 | Screw M5X20 |
| 4 | KZ-4000TP-JT4 | Nut M5 |

# INSPECTION & MAINTENANCE

For safety use, make sure the machine is cleaned every three-months. Before working, make sure to inspect the filter and the nozzle,and ensure there is no impurity. After working, make sure there is no water in the pump and the hose.

If the machine needs to be repaired in the guarantee period, please return the machine to the shop. You need to provide the certificate of purchase in order to repair or get the substitute.

Before Packing, the machine should be cleaned, then use closed packaging, packaging should be firm. Store the machine in dry place.

For transportation safety. The machine should comply with ISTA.



Meaning of crossed - out wheeled dustbin:

Do not dispose of electrical machines as unsorted municipal waste, use separate collection facilities.

If electrical machines are disposed of in landfills or dumps, hazardous substances can leak into the groundwater and get into the food chain, damaging your health and well-being.

When replacing old machines with new ones, the retailer is legally obligated to take back your old machine for disposals at least free of charge.

## Warranty Card

CUSTOMER COPY

Purchaser's Name & Address

Product Name

KZ-4000TP-PRO-2S Tea Pruner

Product Serial No.

Invoice / DN No.

Date of Delivery

Dealer's Office Seal

Dealer Sign

Customer Sign.

I/We accept the terms and conditions of Warranty as in the Warranty Card.

## Warranty Card

DEALER'S COPY

Purchaser's Name & Address

Product Name

KZ-4000TP-PRO-2S Tea Pruner

Product Serial No.

Invoice / DN No.

Date of Delivery

Dealer's Office Seal

Dealer Sign

Customer Sign.

I/We accept the terms and conditions of Warranty as in the Warranty Card.

## Warranty Card

COMPANY COPY

Purchaser's Name & Address

Product Name

KZ-4000TP-PRO-2S Tea Pruner

Product Serial No.

Invoice / DN No.

Date of Delivery

Dealer's Office Seal

Dealer Sign

Customer Sign.

I/We accept the terms and conditions of Warranty as in the Warranty Card.

1. LIMITED WARRANTY PERIOD : For a period of six months from the date of Sale / delivery of the product which ever is earlier. 2. LIMITED WARRANTY OBLIGATION : Replacement of defective part(s) by authorized dealer are free of charges. If found defective by reasons of defective material / poor workmanship, parts will be replaced and there will be no value re-imbursement. 3. TERMS & CONDITIONS : a. Claims under warranty will be accepted on submission of Warranty Card duly filled and stamped by authorized dealer together with the original purchase document. : b. Shipping / transportation / Incidental charges incurred in replacing defective parts under warranty shall be borne by the purchaser.

WARRANTY null & void if

(i) Lack of maintenance, (ii) Incorrect use of the machine or tampering, (iii) Use of non-genuine spare parts, (iv) Usage of acids, solvents, or any flammable materials, (v) Non-usage of filter water / clear water / soft water & using water above 40°C temperature, (vi) While using the pressure washers to draw water which is contaminated with dust & solvents (eg. paint thinners, gasoline, oil, etc.), (vii) Repair done by unauthorized service people, (viii) Those parts subject to wear & tear due to normal operations and plastic parts, rubber items etc, (ix) Loose connections, faulty electrical fixtures & fluctuations in power supply.

4. Should any failure occur during or after the warranty period, customer does not have the right to interrupt payment / price discount.
5. Incidental or consequential damages like inconvenience commercial loss, loss of tie, mental agony etc.
6. Loss of any nature due to operation or non-operation is not covered under warranty.
7. All free service are at Dealer's service point only.
8. Warranty ceases if service is not done periodically.
9. The attached card should be completed and returned to Ratnagiri Impex Pvt. Ltd. for registration immediately after sale / delivery.
10. The warranty is void unless you register the warranty card with Ratnagiri Impex Pvt. Ltd.
11. The decision of Ratnagiri Impex Pvt. Ltd., in settling the warranty is final in all respects.
12. All disputes are subject to Bangalore Courts Jurisdiction

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## 1st Free Service Coupon

ISO 9001-2008 CERTIFIED COMPANY

**FREE SERVICE COUPON (30 Days / 20 hrs (Which ever is earlier)**

Model

**KZ-4000TP-PRO-2S** Tea Pruner

PDI Date

Serial Number

Service No.

Purchase bill no. Date

Dealer's Office Seal & Signature

Customer Signature.....

*Conditions: Operations listed have been performed to my entire satisfaction without any labour charges. Only consumables were charged.*

WARRANTY AND FREE SERVICE COUPON

agriculture | horticulture | sericulture | plantations | forestry | garden | health care

## 2nd Free Service Coupon

ISO 9001-2008 CERTIFIED COMPANY

**FREE SERVICE COUPON (30 Days / 20 hrs (Which ever is earlier)**

Model

**KZ-4000TP-PRO-2S** Tea Pruner

PDI Date

Serial Number

Service No.

Purchase bill no. Date

Dealer's Office Seal & Signature

Customer Signature.....

*Conditions: Operations listed have been performed to my entire satisfaction without any labour charges. Only consumables were charged.*

WARRANTY AND FREE SERVICE COUPON

agriculture | horticulture | sericulture | plantations | forestry | garden | health care

## 3rd Free Service Coupon

ISO 9001-2008 CERTIFIED COMPANY

**FREE SERVICE COUPON (30 Days / 20 hrs (Which ever is earlier)**

Model

**KZ-4000TP-PRO-2S** Tea Pruner

PDI Date

Serial Number

Service No.

Purchase bill no. Date

Dealer's Office Seal & Signature

Customer Signature.....

*Conditions: Operations listed have been performed to my entire satisfaction without any labour charges. Only consumables were charged.*

WARRANTY AND FREE SERVICE COUPON

agriculture | horticulture | sericulture | plantations | forestry | garden | health care


Small Maintenance. Big Savings.


Small Maintenance. Big Savings.


Small Maintenance. Big Savings.




Imported & Marketed by
**RATNAGIRI IMPEX PVT LTD**
Annapurna House, #1/1G, 7th Cross, New Guddadahalli
Mysore Road, Bangalore-560 026
Customer Care: 1800-425-3036
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